A Tale of Two Crises: The Housing Shortage and Homelessness

1 Executive summary

The issue of the housing shortage has been critical for decades; a product of political, social and legislative restrictions, the shortage of housing has reached critical levels. Furthermore, this issue is closely entailed with the problem of homelessness as the restricted supply of housing prevents sufficient affordable housing for those most vulnerable. This report delves deeply into the supply of housing in two distinct U.S. regions: Seattle, Washington, and Albuquerque, New Mexico, the levels of homelessness, and composes them upon each other to create a model informing governments and policymakers and providing solutions to the housing shortage-homelessness problem, not just in the short term, but in the sustainable long term - up to 50 years in the future.

Firstly, we forecasted the housing supplies in both cities utilising linear and exponential regression models, fitting the data points to a linear equation in $y = mx + c$ and and exponential in $y = ae^{bx} + c$ respectively. For Seattle, the linear model predicts the total housing unit increase to 706,423 in 2074, while the exponential model projects a more significant rise to 1,004,600 units by 2074. Albuquerque's growth is more modest, with the linear model forecasting an increase to 324,836 units and the exponential model to 340,102 units by 2074. The exponential model's R^2 value being closer to 1 suggests a better fit, but the finite availability of developable land raises concerns about its long-term applicability, hinting at a possible need for logistic regression models to account for diminishing land availability.

The study extends to predicting changes in homelessness, emphasising rental market dynamics and affordability. In Seattle, multiple linear regression models determining the total housing units in terraces and apartments as well as forecasting the levels of homelessness (which were both extremely suitable models given the R^2 values of 0.945 and 0.899 (3dp) respectively) are superimposed against the potential affordability for homeless individuals, providing an estimate for the number of individuals due to leave homelessness as they work full time. The analysis predicts an increasing gap between affordable housing units and the homeless population unable to afford them (88,632 in 2034, 108,500 in 2044, and 168,106 in 2074), indicating a persistent homelessness challenge despite available housing. Conversely, Albuquerque's homelessness predictions exhibit a complete lack of correlation ($R^2 = 0.042$), suggesting that homelessness there may be driven by factors other than housing affordability, such as substance abuse and mental health issues.

Finally, these previous findings are combined to underscore Seattle's homelessness as primarily an affordability issue, with sufficient housing units but at prices too high for the vulnerable population. Albuquerque's scenario, with a low correlation between homelessness and housing predictions, points towards the need for addressing non-housing-related factors, such as population growth, unemployment rate and poverty. Our comparisons of average salary growth has a negative correlation with poverty, and as the salary increases, people in poverty are projected to fall.

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2 Part I: It Was the Best of Times

2.1 Restatement of the Problem

This question tasks us to predict the housing supply in 2 U.S. regions (Seattle, Washington and Albuquerque, New Mexico) in the next 10, 20 and 50 years, indicating the level of confidence in our predictions.

2.2 Assumptions

1. *There will be no significant policy implementations regarding land usage, fiscal policy, subsidies, social housing investment in Seattle and Albuquerque within the next 10, 20 or 50 years.* As a result of the unpredictability of government enactments or measures, we are unable to account for these potential changes in our predictions. This means that the level of available land for development, the number of construction firms and provision of houses by the government will not fluctuate when they are considered in our prediction.
2. *There will be no local opposition to building on new land in Seattle nor Albuquerque.* The resistance from NIMBYists (not-in-my-backyard: advocates against new developments in urban environments or old neighbourhoods) are losing influence. Their virtual veto power is becoming less significant than ever in governmental decisions in Seattle and Albuquerque so they have not been incorporated in our predictions for housing supply.
3. *The economic environment in Seattle and Albuquerque over the prediction period remains relatively stable.* While economic fluctuations significantly impact housing supply, they are also unforeseeable and incalculable so we have not incorporated them in our model. This means that the rate of the change of price of raw materials and rate of change of the wages of construction workers remains the same over time.
4. *The level of technology and innovation remains constant within the prediction period:* because the data on the level of innovation and technological advancement is limited and would overcomplicate our model, we have chosen not to incorporate it into our prediction.
5. *The demographic of house-buyers remains constant, so the ratio of types of houses remains constant.* The demand for different types of housing would impact the types of housing that become more prevalent in Seattle and Albuquerque. Demographic changes like age distributions, income distributions, family/ household sizes for future generations are so complex that we would not be able to confidently predict them and incorporate that into our model, so we assume that the ratio between detached, semi-detached, terraced houses, and apartment complexes is not prone to fluctuations.

2.3 Model Development

2.3.1 Variable Identification

The supply of housing is composed of 3 factors: the number of inhabited units + the natural level of vacant units + the number of new builds that construction firms are willing and able to construct at a specific price level.

The number of new units built by construction firms are determined by the costs of production. These include costs of labour, land available for development and building materials. Furthermore, government policy, the number of construction companies, technology and innovation contribute to this value. We assumed that these values remained constant for the reasons outlined above.

2.3.2 Linear Regression Model

To predict the supply of housing, we tested two different models, firstly using a linear regression model with Python Scikit-learn, matplotlib, numpy and pandas libraries. This fits our data points to a regression line, in the straight line equation y=mx+c. This model is simple to implement and shows how a response variable, the number of housing units, is affected over a period of time.

The method for this is to consider the explanatory/ independent variable (time) and the response/ dependent variable (the number of housing units), for each data point, generating a linear equation that relates the two, by finding the best values for the two data points to find the best linear fit line. This produces a low error when making predictions and gives a more accurate estimate of model performance. The equation of our linear model is:

Seattle: Total housing units = 6465.14 × (Year-2010) + 292654.02
Albuquerque: Total Housing Units = 1401.18 * (Year- 2010) + 235160.26

2.3.3 Exponential Regression Model

Another model that we used to give our prediction is an exponential regression model, using the same python libraries. We used the same method, generating an exponential equation using the response and explanatory variables and fitting it to the other data points. This yields the exponential equation:

Our x in these equations is equal to (Year - 2010), in order to make our initial year the 0 value of the exponential
X = (Year - 2010)
Seattle: Total housing units =288399*exp(0.0195*x)
Albuquerque: Total housing units = 2540.06 * exp(0.1723 * x) + 234837.38

2.4 Results

We chose to present our linear regression model and exponential regression model on line graphs as below:

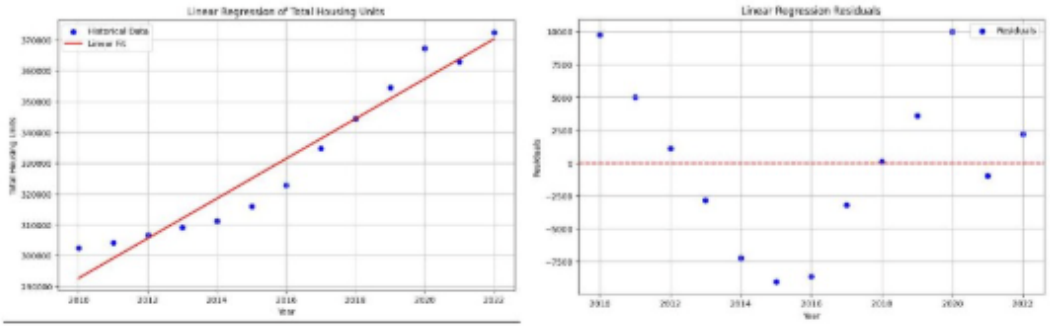


Figure 1: Seattle Linear Regression model and Residuals - Code Ref 1

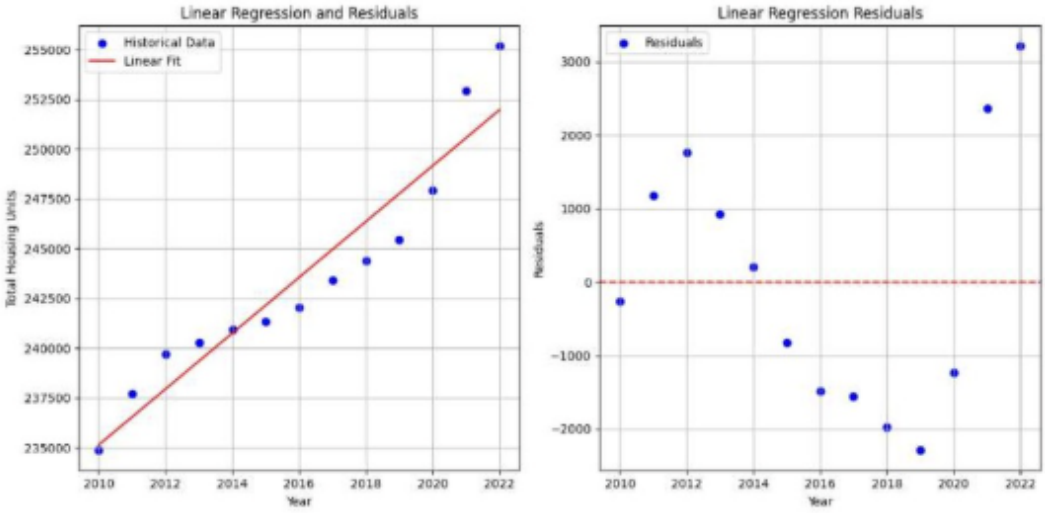


Figure 2: Albuquerque Linear Regression model and Residuals - Code Ref 2

Year	Total housing Units: Seattle (nearest whole number)	Total Housing units: Albuquerque (nearest whole number)
2034	447817	268789
2044	512469	282800
2074	706423	324836

Figure 3: Predictions for the total housing units in Seattle and Albuquerque using a linear regression model

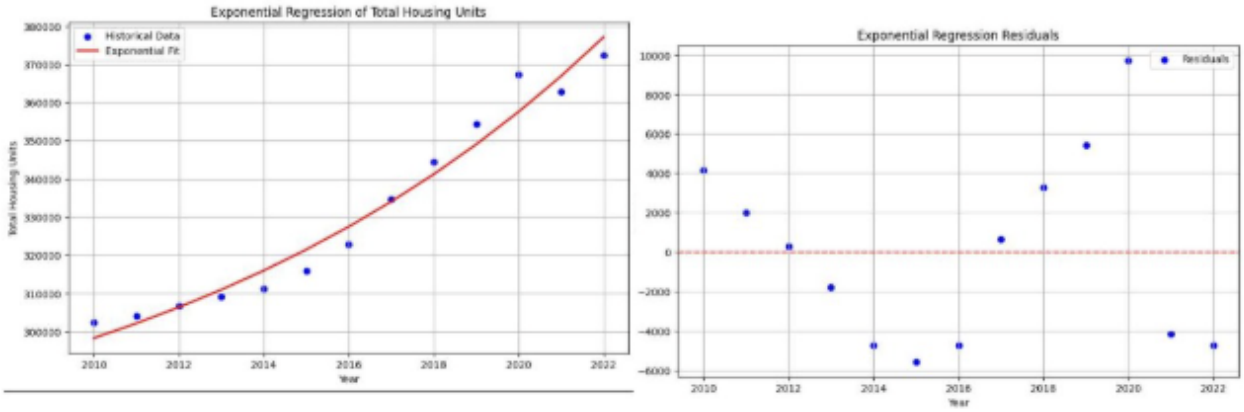


Figure 4: Seattle Exponential regression model and residuals - Code Ref 3

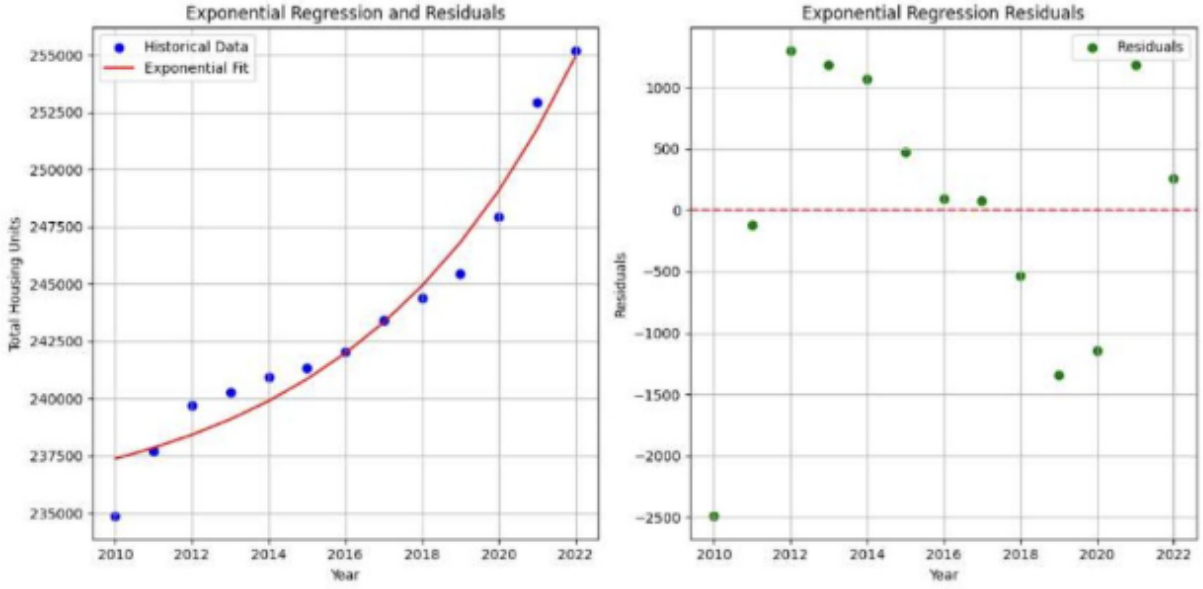


Figure 5: Albuquerque Exponential regression model and residuals - Code Ref 4

Year	Total housing Units: Seattle (nearest whole number)	Total Housing units: Albuquerque (nearest whole number)
2034	460515	270114
2044	559669	280130
2074	1004600	340102

Figure 6: Predictions for total housing units in Seattle and Albuquerque using exponential regression model

2.5 Evaluation

When we used the linear regression model, we generated an R^2 value of 0.942 and 0.905 respectively (3dp.) R^2 is the proportion of the variance of the data that can be explained by our equation. This is derived by the equation:

$$R^2 = 1 - \frac{RSS}{TSS}$$

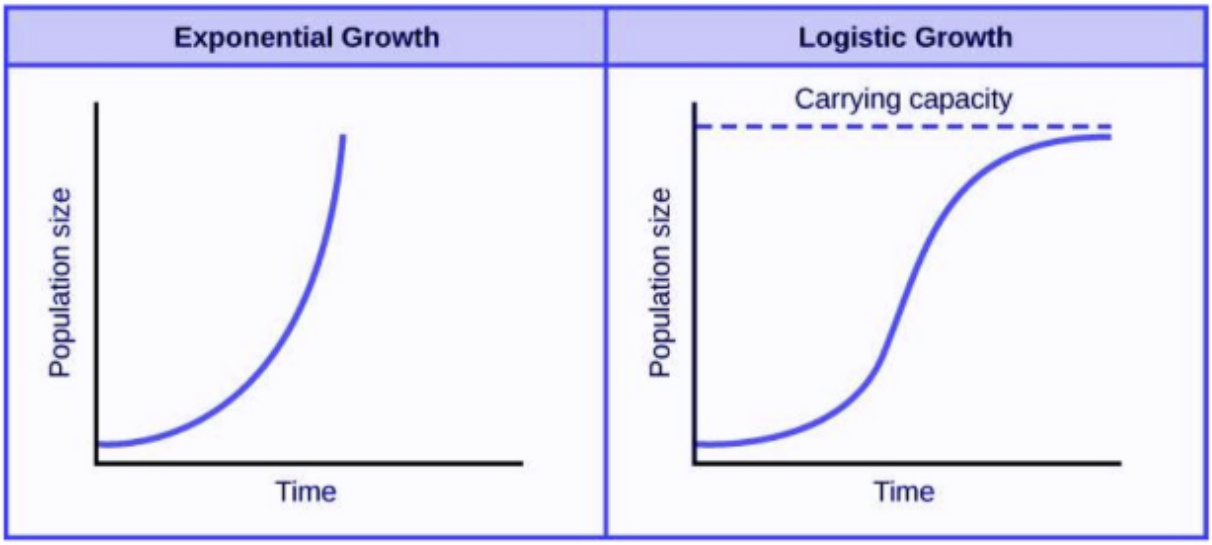
Where RSS is the sum of squares of residuals and TSS is the total sum of squares. Our R^2 variable for our linear regression value is significantly successful, as the closer the value is to one, the better. A sufficient value for R^2 is at least 0.50.

However, the R^2 value when we used an exponential regression model was 0.966 and 0.961 respectively (3dp.) This suggests that an exponential regression model was more successful than the linear regression model.

However, it is important to consider that an exponential model may not be suitable despite this more optimal R^2 value. This is because the availability of spare land available for development remains finite, limiting the supply of housing and causing it to decelerate in the long run - house units will not increase exponentially forever. Due to the availability of data for Seattle, we can calculate a theoretical cap for housing. From this we can imply that there is also a theoretical cap in Albuquerque and demonstrate the unsuitability of an exponential model.

Assuming that the average area of a housing unit in Seattle is 1460 sqft [source] (0.05215 sq miles to 4dp) and the area of Seattle permitted to build houses is 40 miles [source], we divide 40 by 0.05215 to give the theoretical maximum housing in Seattle = 763791. This value is significantly smaller than our 10 year prediction using the exponential model. Alternatively, we could create an interpretable model using FB Prophet, which models time series data as an additive combination of trend, seasonality and noise components which is suitable for time series in which seasonal variation is consistent.

This hard cap on the amount of homes in Seattle would suggest that a logistic regression model may have been more suitable. This would include in our model a factor which decreases the rate of change closer to the theoretical maximum our model predicts. This would be more accurate because as the available land space decreases, this would increase the cost of building on this land, leading to a reduced rate of production. However, we were unable to capture any of this future decrease in the rate of building construction, because in our data at this time it is not able to be seen as a significant factor yet. Thus, we can guarantee that our results for longer time progressions (such as 2074) would be inaccurate, but given the data we have we are unable to quantify this with real numbers.



We could have estimated where we are on the logistic curve[10] by representing the current building count as a proportion of the carrying capacity, which would result in us being able to estimate where we are in the progression. Unfortunately, the level of confidence we could have in this is very low, meaning that we cannot build this into our model. Given further time or data, we would estimate this and create a more accurate model.

2.6 Final prediction and confidence of results

In our answer to this section, we have considered our linear regression result as the minimum value we could predict the amount of housing units there could be, and the exponential regression as the maximum value to predict, giving us an interval for our results

Seattle:

10 Year prediction interval : 447817 (linear) - 460515 (exponential) ~ 12698 interval
20 Year prediction interval: 512469 (linear) - 559669 (exponential) ~ 47200 interval
50 Year prediction interval: 706423 (linear) - 1004600 (exponential) ~ 298177 interval

Our 10 Year prediction we can be fairly confident in due to the small range as a proportion of the total, and we would estimate that the true amount of houses constructed to be top of this range

Our 20 Year prediction we can also be fairly confident in, but we would assume that due to the housing cap becoming an effect in this result, we would expect the true number of houses constructed to be nearer the middle of this range

Our 50 Year prediction unfortunately we cannot be very confident in, as the bottom of the range is very close to the absolute housing cap we have estimated, combined with the huge range as a proportion of the total (nearly 30% of this answer), resulting in this answer not likely to be a true representation of the number of buildings constructed

Albuquerque:

10 Year prediction interval : 268789 (linear) - 270114 (exponential) ~ 1325 interval
20 Year prediction interval: 282800 (linear) - 280130 (exponential) ~ 2670 interval
50 Year prediction interval: 324836 (linear) - 340102(exponential) ~ 15266 interval

Immediately, it is noticed that these predictions have much smaller ranges than Seattle.

Our 10 year prediction has a small range, indicating we can be confident in the results this produces, but we would expect the results to be close to the lower bound of these results, as the linear term more accurately describes the growth in these 10 years

Our 20 year prediction also has a small range, however for this result the linear trend actually predicts higher than the exponential trend, showing there may be an inaccuracy in the model. Nonetheless, we could predict with confidence that the result would lie in this interval, again nearer the linear calculation

Our 50 year prediction has a larger range, meaning we could be less confident in the exact answers produced, however as the range is still a small proportion of the total, we can still have relative confidence. Obviously, predicting any further into the future is likely to be inaccurate, as the exponential term will grow to infinity, but for only 50 years, it appears to be accurate enough.

3 Part II: It Was the Worst of Times

3.1 Restatement of the problem

This question tasks us to predict the changes in the homeless population in the next 10, 20 and 50 years.

3.2 Assumptions

1. *Everyone who is at risk to homelessness will be renting homes, so only data on renting (e.g. renting prices per month) will be considered.* This is because banks that distribute mortgages will run credit checks and assess one's suitability for a mortgage, so it is unlikely that house prices will be an adequate indicator of poverty/ homelessness.
2. *Everyone at risk to homelessness will only look at renting terraces and apartments (in which the number of units per plot is at least 3-4 assuming the price of 3-4 unit plots is uniformly distributed, this would be at least 3.5 units, the midpoint of the data range. This is applied to every single midpoint seen in the data provided), and not houses.* This is because apartment rent per is significantly cheaper than house rent per month - this is because apartment complexes are generally bought by corporations due to economies of scale in which the cost per unit is lower; there is no incentive for the individual buy a more expensive property when their income is so low, so we did not incorporate the few that make this economic decision in our prediction.

3. *When in desperate circumstances, an individual will spend a maximum of 60% of their income on rent.* This was calculated by (1 – proportion of spending on vital needs).
4. *Of the homeless people that can afford the minimum rent price of a house, they haven't yet found it on the market, but soon will* - this is because the availability of houses that are this price are very low.
5. *Homeless people live alone and buy houses alone*. This is because the people that live alone do not have as much eligibility for benefits and do not enjoy a joint salary with a partner.
6. *The rate of inflation remains constant over time.* This is because we are assuming a stable economy and any fluctuations are incalculable.

3.3 Model Development

The determinants of homelessness in Seattle is combined in a manipulation of multiple linear regression models to determine how many homeless people can potentially afford a house:

The total number of units in every terrace and apartment. This is the availability of units that an individual that is homeless would potentially buy, variable Y - this can also be considered the latent demand from homeless people (what they are willing to buy.) This value is then predicted for 10, 20 and 50 years in the future using a linear regression model. This was for the same reasons listed in 2.3.2. Using the same method also, we yielded the equation:

$$Y = 100593.05 * year – 200111883.08$$

Using the total availability of housing units in terraces and apartments, we must calculate the number of these that could potentially be affordable by a homeless person. We did this by multiplying this by 0.02; this 2% value is derived from the lower bound of apartment rents in Seattle. This means that the only 2% of rent for apartments in Seattle are between $700 and $1000 [source].

Next, we need to calculate the minimum income of homeless people to afford this housing and compare it to the number of homeless people's income that exceed this number to find the true value of homeless people. This is because we are assuming that homeless people that can afford the cheapest terraces and apartments will eventually find them. Using the assumption that in desperate circumstances, the minimum proportion of income that will cover vital needs like food and water is 40%. This [source] claims that 50% of income should be spent on necessities, but we allowed a 10% margin, restricting expenditure to pure necessities:

12 * (1000/0.6) = $20,000, the minimum salary required to rent the cheapest apartments.

In tandem with this, we used another linear regression model to predict the total number of homeless people (H) 10, 20 and 50 years from now for the same reasons, yielding the following equation:

$$H = 192.62 * year – 378261.45$$

We assume that the number of homeless people whose income exceeds the minimum of $20,000 is the number of homeless people working full-time. 7.5% of the homeless report working full time [source], and we estimate that these people stop being homeless within time. The difference between the predictions for homelessness and the number of people who will soon stop being homeless because they work fulltime is our prediction for the true trend of homelessness in our prediction period.

3.4 Results

3.4.1 Seattle

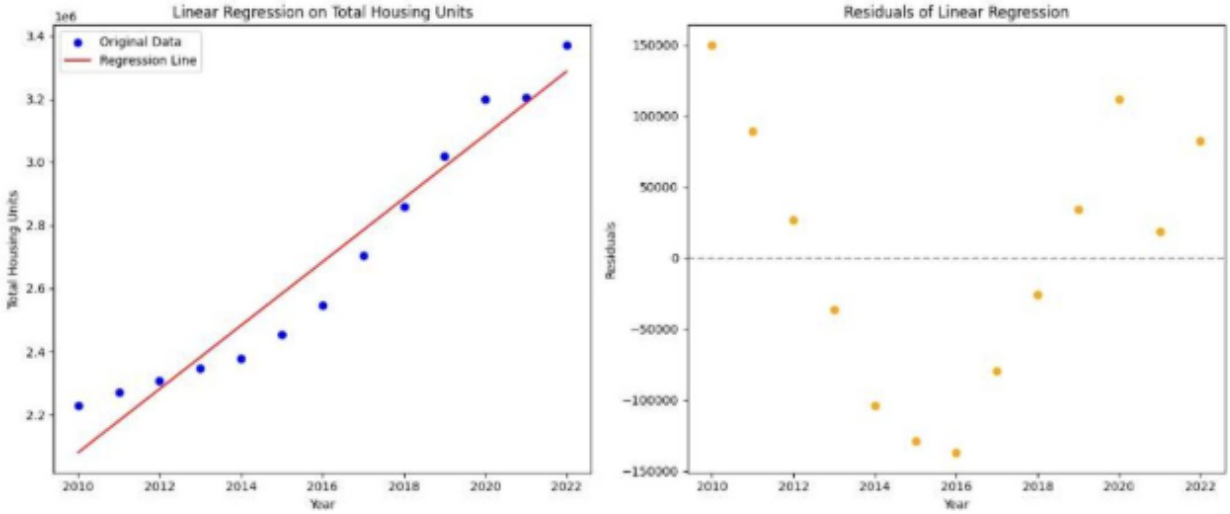


Figure 7: Linear regression model of total housing units (the total available units that one individual could inhabit) in Seattle - Code Ref 5

$$R^2 = 0.945294662509853$$

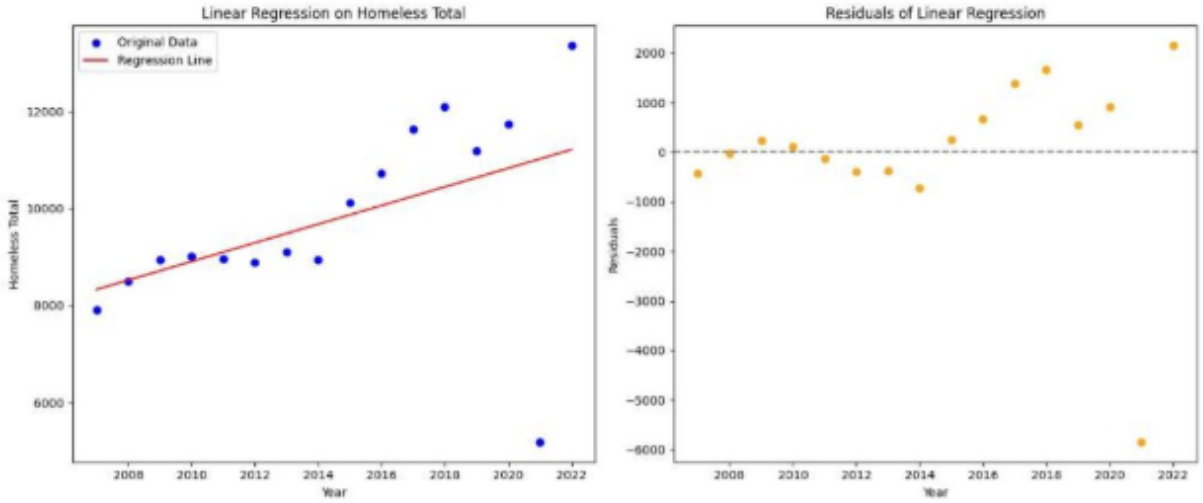


Figure 8: Linear regression model of total homeless people in Seattle - Code Ref 6

$$R^2 = 0.8993164018811541$$

The results for each step of our modelling method and computation is shown below, with a method shown for our 2034 predictions.

	2034	2044	2074
Total housing units in terraces and apartments (nearest whole)	4,494,391	5,500,321	8,518,113
Total affordable units (nearest whole)	4,494,391 * 0.02 = 89888	110,006	170,362
Total homeless (nearest whole)	16,743	20,080	30,089
Total homeless that can afford 'affordable' homes (nearest whole)	16,743 * 0.075 = 1,256	1,506	2,256
Total homeless that can't afford 'affordable' homes (nearest whole)	16,743 * (1-0.075) = 88,632	108,500	168,106

Furthermore, the number of homeless people that can afford 'affordable' housing units < the availability of 'affordable' housing we yielded, 20,000.

3.4.2 Albuquerque

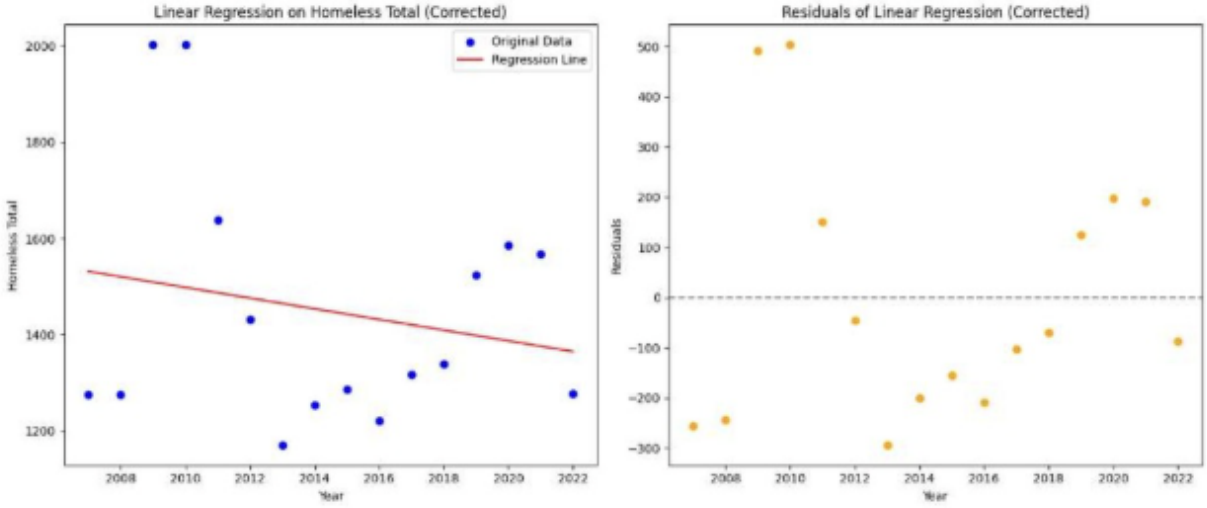


Figure 8: Linear Regression of Total Homeless People

$R^2 = 0.042023346395244854$

Mean = 1455.31
Standard deviation = 259.01

Therefore, the levels of homelessness will vary within a range of one standard deviation on both sides of the mean to the nearest whole:

$$1196 < \text{mean homelessness in albuquerque} < 1714$$

The reasons for this will be explained shortly in the 3.5 - Evaluation and Discussion section.

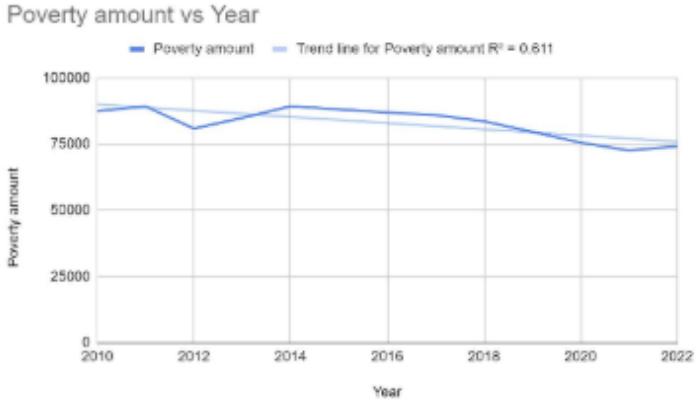
3.5 Evaluation and Discussion

The predicted true homeless population in Seattle was significantly more suitable, because we achieved an R^2 value of 0.945 (3dp) for the total housing units and 0.899 for the total number of homeless people. Being extremely close to 1 this is a sufficient prediction of total housing and homelessness respectively. Furthermore, we were also able to determine that the number of homeless people that can afford 'affordable' housing units is less than the availability of 'affordable' housing we yielded, 20,000. This suggests that the homelessness issue in Seattle is a product of affordability issues and not availability issues. There are enough housing units but the market equilibrium price of houses for rent is still too high for those most vulnerable. This suggests that solutions to homelessness are to potentially increase the supply of housing in order to put deflationary pressure on the housing market. We will explore this further in 4: Rising from This Abyss. Our values of the absolute number of homeless people grow slowly, growing to roughly 30,000 in 2074, which could show we had been accurate.

However, our predictions for the true homeless population in Albuquerque is significantly less suitable, with an R^2 value of 0.042 (3dp). This suggests that there is no trend between rising homelessness in Albuquerque. Potentially, the causes of homelessness in Albuquerque are unrelated to their affordability of housing in that area, and in fact could be a result of exogenous factors such as drugs, mental health disorders, and addiction.

3.6 An alternate Method?

In an attempt to raise higher confidence intervals, we created a second method for forecasting the amount of homeless people in Seattle. Based on the random nature of the count of homeless people in Albuquerque, the new model has not been seen to be relevant.



We noticed there is a negative correlation between the amount of people in Poverty and time in Seattle, and calculated the graph. There is also a correlation between the amount of people in poverty and the amount that are homeless, however due to time concerns, we were not able to complete the analysis of this. This model, however, would have given us a lower bound for the amount of people that were homeless, allowing a sense of scale of our results to be derived. This would be something that we could complete further given time, but unfortunately at this time frame we were unable to do this. Tentatively, we can say that we believe the model to show an increasing population of homeless in Seattle, levelling off in the future, due to decreasing levels of people in poverty over time.

4 Part III: Rising from This Abyss

4.1 Restatement of the problem

This question tasks us with creating a model that would help Seattle determine a long term plan to address homelessness. We are also asked to consider how adaptable our model is to unforeseen circumstances like natural disasters, economic recessions or increased migrant populations.

4.2 Assumptions

1. *The relationship between the independent variable (year) and the dependent variables (Average Salary, Percentage, Population, and People in Poverty) is assumed to be linear.* This captures the basic trend and its strength, which. This means that a change in the independent variable is expected to result in a proportional change in the dependent variable.
2. *Homoscedasticity: The variance of error terms is consistent across all levels of the independent variable.* This means that the residuals are spread evenly across the regression line, and there is no pattern in the spread of residuals.
3. *The relationships between the independent and dependent variables have remained stable over the period in question and will continue to do so in the future*. As previously stated, this is a product of the unpredictability of these events that may trigger fluctuations in the dependent variable.

4.3 Model Development

4.3.1 Decision Tree Regressor

We first decided to develop a Decision Tree Regressor, which looks at all the features- Average salary, Percentage, Population- to predict people in poverty, by splitting the data through a series of decisions, creating smaller and smaller groups, and the final prediction is based on the average values of the smallest group.

We first fitted the features data to the data we want to predict (people in poverty), which will create a decision tree that can be applicable for forecasting people in poverty

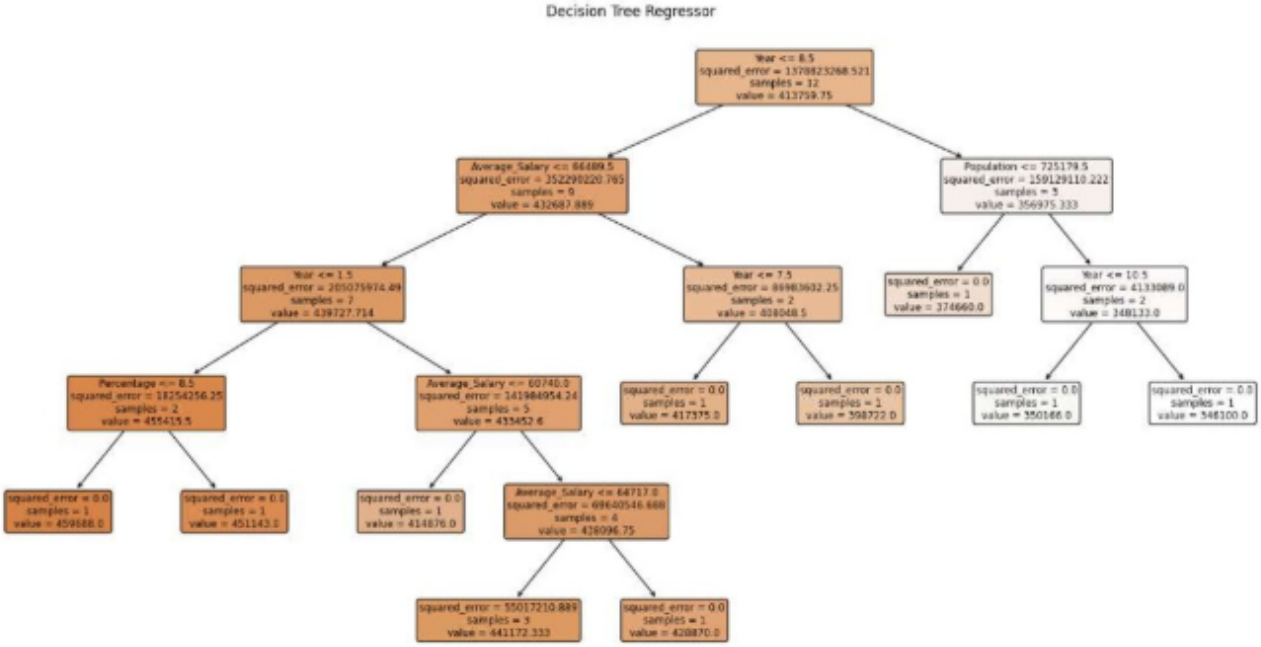


Figure 9: Decision tree regressor

This however came with a couple of problems. (1) Decision Tree Regressors require multiple data points to provide accurate results, so using a small dataset results in overfitting- when a model is too tailored to its training dataset, it will not be effective for external data that doesn't fit the trends followed in the training data. Furthermore, there is no way to evaluate the accuracy of the model, since we don't have enough data points to form a training set. 2. To make an accurate prediction, we would have to input the forecasted values for features, e.g. to predict people in poverty in 2030, we need the forecasted features as well- the population in 2030. This makes the model useless unless we obtain the forecasted values. If we were to have more data points, we still wouldn't use a Decision Tree Regressor, and instead use a time series method such as Single Exponential Smoothing or Triple Exponential Smoothing if our data contained seasonality.

4.3.2 Linear Regression

To resolve this issue, we decided to carry out multiple linear regressions for each feature against time, as well as a linear regression for people in poverty against time, then compared the trends of the features against the trend of people in poverty over time.

1. Homelessness is mostly affected by poverty, so we will be finding whether or not external factors, such as population increase or education affect poverty.
2. People in poverty will be calculated by population multiplied by poverty rate
3. We collected factors that we believed would affect poverty - average salary, unemployment percentage and population growth, then compare it to the rate of people in poverty
4. We then carried out separate linear regressions for each variable, then compared it to the rate of people in poverty.

4.4 Results

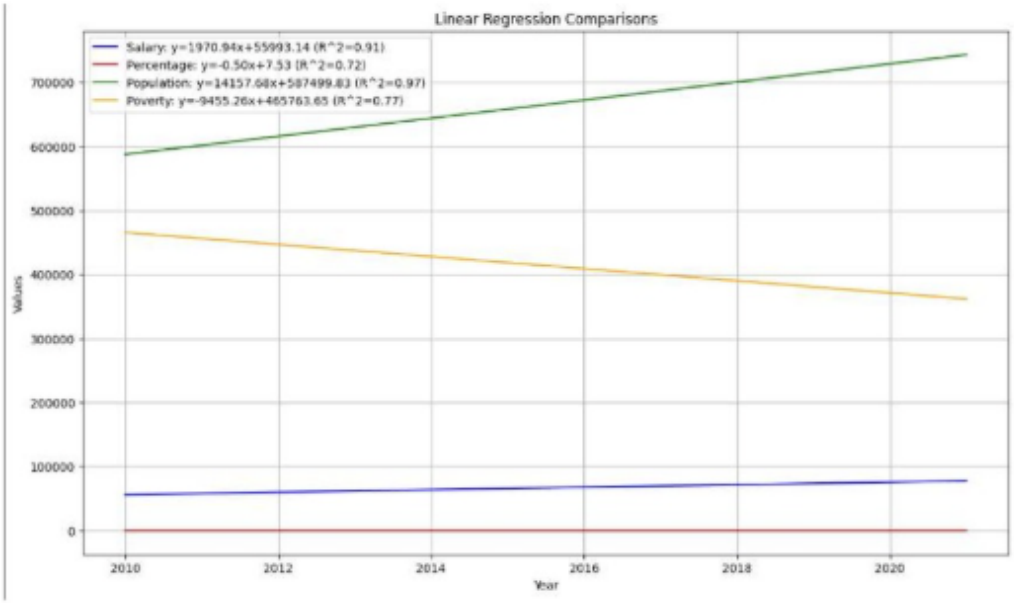
The provided code performs linear regression analyses on four different datasets, each representing a different variable over the years 2010 to 2021: average salary, percentage (of unemployment), population, and people in poverty. The analyses yield linear models that capture the trends in each dataset over time.

The linear regression model for average salary shows a consistent increase in salary over the years, with an annual increase rate represented by the slope of 1970.94 and a starting value (intercept) of 55993.14 in 2010. The model has a high R^2 value of 0.97, indicating a strong fit to the data, suggesting that the average salary has steadily increased over the years.

The percentage dataset's linear model, with a slope of -0.50 and an intercept of 7.53, suggests a gradual decline over the years. The R^2 value of 0.88 indicates a good fit, meaning the variable represented by "percentage" has been decreasing annually by about 0.5%.

Population data analysis reveals an increasing trend, with the population growing by approximately 14157.68 individuals each year, starting from a base of 587499.83 in 2010. The model's R^2 value of 0.97 demonstrates a very strong fit, indicating a consistent population growth during the analysed period.

The people in poverty dataset presents a declining trend, with a slope of -9455.26, indicating a reduction in the number of people in poverty by approximately 9455 individuals annually, starting from an initial count of 465763.65 in 2010. The model's R^2 value of 0.92 suggests a strong correlation and a good fit to the data, reflecting a notable decrease in poverty levels over the years.



4.5 Evaluation

The evaluation of the linear regression analyses conducted on the datasets for average salary, percentage, population, and people in poverty from 2010 to 2021 reveals several insights and considerations.

Strengths:

Model Fit: The high (R^2) values for all models (ranging from 0.88 to 0.97) indicate that linear regression provides a strong fit to the datasets, explaining a significant portion of the variance in each dependent variable based on time alone.

Simplicity and Interpretability: The linear models are straightforward, making them easy to understand and interpret. The coefficients directly indicate the rate of change in the dependent variables per year, providing clear insights into trends.

Predictive Power: The models offer predictive capabilities, enabling forecasts of future values based on the established trends, which can be valuable for planning and decision-making purposes.

Limitations:

Linearity Assumption: The models assume a linear relationship between the year and each variable. While this assumption holds well within the observed time frame, it may not capture nonlinear trends or fluctuations that could occur outside this period.

Oversimplification: By focusing solely on the year as the independent variable, the models may oversimplify complex phenomena. Factors other than time, such as economic policies, demographic changes, or external events, might influence the trends but are not accounted for in these models.

Extrapolation Risk: Using these models to make predictions far into the future carries risks, as the linear trends established over the past decade may not persist indefinitely. Long-term forecasts could be inaccurate if the underlying dynamics change.

Potential for Overfitting: While not evident from the high R^2 values, there's always a risk of overfitting, especially if the models were to include more predictors. Overfitting can make the models less generalizable to unseen data.

Recommendations for Future Research:

Incorporate Additional Variables: Future analyses could benefit from including more independent variables to capture the influence of other factors on the trends observed.
Explore Non-Linear Models: Investigating non-linear relationships or models that can capture more complex dynamics might provide a more nuanced understanding of the trends.

Validate Predictions: Where possible, predictions made by these models should be validated with actual data as it becomes available to assess the models' accuracy over time.

The adaptability of linear regression models to unforeseen circumstances is a critical factor to consider, especially when these models are used for forecasting future trends based on historical data. While linear regression models offer simplicity and interpretability, their adaptability to sudden, unexpected changes in the underlying trends can be limited. This limitation primarily stems from the core assumptions and characteristics of linear regression:

Adaptability in Practice:

Re-training with New Data: The adaptability of a linear regression model can be somewhat improved by periodically re-training the model with the most recent data. This approach can help the model to 'catch up' with recent trends but may not be timely enough to account for abrupt changes.

Incorporating External Indicators: Enhancing the model to include external factors or leading indicators can increase its responsiveness to changes. However, identifying and obtaining reliable indicators can be challenging, and the model's complexity and interpretability may be affected.

Hybrid or Dynamic Models: Combining linear regression with other modelling techniques can improve adaptability. Machine learning models that can update their parameters in real-time based on new data might also offer more flexibility.

Linear regression models are valuable for analysing historical data and identifying trends based on past patterns. However, their effectiveness is limited when facing unforeseen circumstances due to their static nature and reliance on historical data. These models assume linear relationships and constant conditions over time, making them less adaptable to sudden changes or external shocks.

Advanced modelling techniques, like time series analysis and machine learning, offer improved adaptability by accommodating dynamic changes and non-linear relationships. Incorporating external variables and continuously updating and validating models can enhance their resilience to unexpected events. Decision trees, as an example of machine learning, provide a more flexible approach by branching out scenarios based on various conditions, allowing for the consideration of a wider range of factors and potential outcomes.

In essence, overcoming the limitations of linear regression in the face of unpredictability requires the integration of more sophisticated, dynamic modelling approaches and a commitment to ongoing model refinement.

5 Conclusion

The report highlights the complexity of housing and homelessness crises, with distinct challenges and solutions in different regions. For Seattle, increasing the supply of affordable housing emerges as a key strategy, while Albuquerque requires a multifaceted approach addressing broader social issues. Future models should consider the limitations of available land and incorporate logistic regression or FB Prophet to better predict housing supply constraints.
We created both linear and exponential regression methods to predict the volume of housing that would be available in the future, providing lower and upper bounds to compare.

5.1 Further Studies

The data presented in this report gives a clear indication that further data is required to solve such a complex issue. This issue has resulted in many of our results possibly being inaccurate, due to the difficulty of extrapolating large periods of time from small datasets. The predictions for smaller timescales remain accurate, however our long term predictions cannot possibly be taken to truth due to how far they stray and the nature of world events mean that situations are likely to have changed that far in the future. In the future, being able to analyse the effects of more variables on our results would be preferable.

5.2 Summary

Our first model produces two regression coefficients for the sets of data of Seattle and Albuquerque, and computes the upper and lower bounds of the volume of plots that could be constructed. This resulted in much smaller bounds being created for Albuquerque than Seattle, showing us that we could be more confident in those results. However, due bounds being created for results across the board, this can allow us to get an accurate depiction of the amount of potential new plots . To forecast the growth of homeless population, we used two regression models to predict the total available housing units that will be available to rent, since they will be the most likely type of housing that is available to the homeless, then apply the lower bound of rent prices to the available housing units, to grasp the true number of available houses to rent. By applying a ratio of homeless people that work full time, we can then see which percentage of the homeless can afford to buy an available unit, and we then subtract the total amount of homeless people that can afford a unit from the previously projected homeless population growth. Given further time, we could have calculated lower bounds for these values, but we did not have time in this occasion
In our third model, we plotted four linear regressions on one set of axes to show a metric of poverty, which we believed would help the city of Seattle to reduce homelessness throughout the city. From our research, we deemed that poverty was the greatest contributor to homelessness, hence this was our main metric that we took into consideration.

6 References

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6.1 Part I: It Was the Best of Times

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6.2 Part II: It Was the Worst of Times

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7 Technical Code Reference

Seattle Linear Regression of Total Housing Units - Code Ref 1

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Provided data
data = {
    "Year": [2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    "Total housing units": [302465, 304164, 306694, 309205, 311286, 315950, 322795, 334739, 344583, 354475, 367337, 362809, 372436]
}

# Create a DataFrame
df = pd.DataFrame(data)

# Prepare the data for modeling
X = df[["Year"]]
y = df["Total housing units"]

# Instantiate and fit the model
model_linear = LinearRegression()
model_linear.fit(X, y)

# Calculate R^2 value
r_squared_linear = model_linear.score(X, y)

# Equation of the line
slope_linear = model_linear.coef_[0]
intercept_linear = model_linear.intercept_
equation_linear = f"y = {slope_linear:.2f} * x + {intercept_linear:.2f}"

# Plotting
plt.figure(figsize=(10, 6))
plt.scatter(X, y, color='blue', label='Historical Data')
plt.plot(X, model_linear.predict(X), color='red', linewidth=2, label='Linear Fit')
plt.xlabel('Year')
plt.ylabel('Total Housing Units')
plt.title('Linear Regression of Total Housing Units')
plt.legend()
plt.grid(True)
plt.show()
print(r_squared_linear)
print(equation_linear)
```

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Seattle Residuals of Linear Regression - Code Ref 6

```python
# Linear Regression Predicted values
y_pred_linear = model_linear.predict(X)

# Linear Regression Residuals
residuals_linear = y - y_pred_linear

# Plotting Linear Regression Residuals
plt.figure(figsize=(10, 6))
plt.scatter(X, residuals_linear, color='blue', label='Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel('Year')
plt.ylabel('Residuals')
plt.title('Linear Regression Residuals')
plt.legend()
plt.grid(True)
plt.show()
```

Seattle Exponential Regression of Total Housing Units - Code Ref 3

```
from scipy.optimize import curve_fit

# Define the exponential function to fit
def exponential_func(x, a, b, c):
    return a * np.exp(b * (x - 2010)) + c

# Fit the exponential model
params_exp, covariance_exp = curve_fit(exponential_func, df['Year'], df['Total housing units'])

# Predicted values using the fitted exponential model
y_pred_exp = exponential_func(df['Year'], *params_exp)

# Calculate R^2 for the exponential model
ss_res_exp = np.sum((df['Total housing units'] - y_pred_exp) ** 2)
ss_tot_exp = np.sum((df['Total housing units'] - np.mean(df['Total housing units'])) ** 2)
r_squared_exp = 1 - (ss_res_exp / ss_tot_exp)

# Equation of the fitted exponential line
equation_exp = f"y = {params_exp[0]:.2f} * exp({params_exp[1]:.4f} * (x - 2010)) + {params_exp[2]:.2f}"

# Plotting the exponential regression results
plt.figure(figsize=(10, 6))
plt.scatter(df['Year'], df['Total housing units'], color='blue', label='Historical Data')
plt.plot(df['Year'], y_pred_exp, color='red', linewidth=2, label='Exponential Fit')
plt.xlabel('Year')
plt.ylabel('Total Housing Units')
plt.title('Exponential Regression of Total Housing Units')
plt.legend()
plt.grid(True)
plt.show()
print(r_squared_exp)
print(equation_exp)
```

Seattle Residuals of Exponential Regression - Code Ref 5

```
# Exponential Regression Residuals
residuals_exp = df['Total housing units'] - y_pred_exp

# Plotting Exponential Regression Residuals
plt.figure(figsize=(10, 6))
plt.scatter(df['Year'], residuals_exp, color='blue', label='Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel('Year')
plt.ylabel('Residuals')
plt.title('Exponential Regression Residuals')
plt.legend()
plt.grid(True)
plt.show()
```

Albuquerque Linear Regression, Exponential Regression, and their residuals - Code Ref 2,4

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit

# New provided data
data_new = {
    "Year": [2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    "Total housing units": [234891, 237735, 239718, 240277, 240961, 241326, 242070, 243402, 244382, 245476, 247926, 252924, 255178]
}

# Create a new DataFrame
df_new = pd.DataFrame(data_new)

# Prepare the data for modeling
X_new = df_new[["Year"]]  # Predictor variable
y_new = df_new["Total housing units"]  # Response variable

# Linear Regression Model
model_linear_new = LinearRegression()
model_linear_new.fit(X_new, y_new)
y_pred_linear_new = model_linear_new.predict(X_new)

# Exponential Regression Model
def exponential_func(x, a, b, c):
    return a * np.exp(b * (x - 2010)) + c

params_exp_new, _ = curve_fit(exponential_func, df_new['Year'], df_new['Total housing units'])
y_pred_exp_new = exponential_func(df_new['Year'], *params_exp_new)

# Calculating R^2 values
r_squared_linear_new = model_linear_new.score(X_new, y_new)
r_squared_exp_new = 1 - np.sum((y_new - y_pred_exp_new) ** 2) / np.sum((y_new - np.mean(y_new)) ** 2)

# Calculating Residuals
residuals_linear_new = y_new - y_pred_linear_new
residuals_exp_new = y_new - y_pred_exp_new

# Equation of the lines
equation_linear_new = f"y = {model_linear_new.coef_[0]:.2f} * x + {model_linear_new.intercept_:.2f}"
equation_exp_new = f"y = {params_exp_new[0]:.2f} * exp({params_exp_new[1]:.4f} * (x - 2010)) + {params_exp_new[2]:.2f}"
```

continued below

```python
# Plotting Linear Regression with Residuals
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.scatter(X_new, y_new, color='blue', label='Historical Data')
plt.plot(X_new, y_pred_linear_new, color='red', label='Linear Fit')
plt.title('Linear Regression and Residuals')
plt.xlabel('Year')
plt.ylabel('Total Housing Units')
plt.legend()
plt.grid(True)
print(equation_linear_new)
print(r_squared_linear_new)

plt.subplot(1, 2, 2)
plt.scatter(X_new, residuals_linear_new, color='blue', label='Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.title('Linear Regression Residuals')
plt.xlabel('Year')
plt.ylabel('Residuals')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

# Plotting Exponential Regression with Residuals
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.scatter(df_new['Year'], df_new['Total housing units'], color='blue', label='Historical Data')
plt.plot(df_new['Year'], y_pred_exp_new, color='red', label='Exponential Fit')
plt.title('Exponential Regression and Residuals')
plt.xlabel('Year')
plt.ylabel('Total Housing Units')
plt.legend()
plt.grid(True)
print(equation_exp_new)
print(r_squared_exp_new)

plt.subplot(1, 2, 2)
plt.scatter(df_new['Year'], residuals_exp_new, color='green', label='Residuals')
plt.axhline(y=0, color='red', linestyle='--')
plt.title('Exponential Regression Residuals')
plt.xlabel('Year')
plt.ylabel('Residuals')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

Seattle's Linear regression of total calculated housing units and residuals

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Data
housing_data = {
    'Year': [2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Total_Housing_Units': [2229814, 2270187, 2307873, 2345393, 2378749, 2453830, 2546693, 2704416, 2858934, 3019705, 3197955, 3204095, 3369760]
}

# Convert data to DataFrame
housing_df = pd.DataFrame(housing_data)

# Define features and target variable
X_housing = housing_df[['Year']]
y_housing = housing_df['Total_Housing_Units']

# Create and fit the linear regression model
model_housing = LinearRegression()
model_housing.fit(X_housing, y_housing)

# Predictions
predictions_housing = model_housing.predict(X_housing)

# Equation of the line
slope_housing = model_housing.coef_[0]
intercept_housing = model_housing.intercept_
equation_housing = f"Total Housing Units = {slope_housing:.2f} * Year + {intercept_housing:.2f}"

# R^2 value
r_squared_housing = model_housing.score(X_housing, y_housing)

# Calculate residuals
residuals_housing = y_housing - predictions_housing

# Plotting the regression and residuals
plt.figure(figsize=(14, 6))

# Plot 1: Linear Regression
plt.subplot(1, 2, 1)
plt.scatter(X_housing, y_housing, color='blue', label='Original Data')
plt.plot(X_housing, predictions_housing, color='red', label='Regression Line')
plt.title('Linear Regression on Total Housing Units')
plt.xlabel('Year')
plt.ylabel('Total Housing Units')
plt.legend()
```

```python
# Plot 2: Residuals
plt.subplot(1, 2, 2)
plt.scatter(X_housing, residuals_housing, color='orange')
plt.axhline(y=0, color='gray', linestyle='--')
plt.title('Residuals of Linear Regression')
plt.xlabel('Year')
plt.ylabel('Residuals')

# Show plots
plt.tight_layout()
plt.show()

# Output the equation, R^2, and residuals
print("Equation of the line:", equation_housing)
print("R^2 value:", r_squared_housing)
print("Residuals:", residuals_housing.tolist())
```

Seattle's Linear regression on Homeless Total and residuals

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Homeless total data
homeless_data = {
    'Year': [2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Homeless_Total': [7902, 8501, 8952, 9022, 8972, 8899, 9106, 8949, 10122, 10730, 11643, 12112, 11199, 11751, 12560, 13368]
}

# Convert data to DataFrame
homeless_df = pd.DataFrame(homeless_data)

# Define features and target variable
X_homeless = homeless_df[['Year']]
y_homeless = homeless_df['Homeless_Total']

# Create and fit the linear regression model
model_homeless = LinearRegression()
model_homeless.fit(X_homeless, y_homeless)

# Predictions
predictions_homeless = model_homeless.predict(X_homeless)

# Equation of the line
slope_homeless = model_homeless.coef_[0]
intercept_homeless = model_homeless.intercept_
equation_homeless = f"Homeless Total = {slope_homeless:.2f} * Year + {intercept_homeless:.2f}"

# R^2 value
r_squared_homeless = model_homeless.score(X_homeless, y_homeless)

# Calculate residuals
residuals_homeless = y_homeless - predictions_homeless

# Plotting the regression and residuals
plt.figure(figsize=(14, 6))

# Plot 1: Linear Regression
plt.subplot(1, 2, 1)
plt.scatter(X_homeless, y_homeless, color='blue', label='Original Data')
plt.plot(X_homeless, predictions_homeless, color='red', label='Regression Line')
plt.title('Linear Regression on Homeless Total')
plt.xlabel('Year')
plt.ylabel('Homeless Total')
plt.legend()
```

```python
# Plot 2: Residuals
plt.subplot(1, 2, 2)
plt.scatter(X_homeless, residuals_homeless, color='orange')
plt.axhline(y=0, color='gray', linestyle='--')
plt.title('Residuals of Linear Regression')
plt.xlabel('Year')
plt.ylabel('Residuals')

# Show plots
plt.tight_layout()
plt.show()

# Output the equation, R^2, and residuals
print("Equation of the line:", equation_homeless)
print("R^2 value:", r_squared_homeless)
print("Residuals:", residuals_homeless.tolist())
```

Albuquerque's Linear Regression on Homeless Total and residuals

```python
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Corrected homeless total data
homeless_data_corrected = {
    'Year': [2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022],
    'Homeless_Total': [1276, 1276, 2002, 2002, 1639, 1431, 1171, 1254, 1287, 1222, 1318, 1340, 1524, 1586, 1567, 1277]
}

# Convert corrected data to DataFrame
homeless_df_corrected = pd.DataFrame(homeless_data_corrected)

# Define features and target variable for the corrected data
X_homeless_corrected = homeless_df_corrected[['Year']]
y_homeless_corrected = homeless_df_corrected['Homeless_Total']

# Create and fit the linear regression model for the corrected data
model_homeless_corrected = LinearRegression()
model_homeless_corrected.fit(X_homeless_corrected, y_homeless_corrected)

# Predictions for the corrected data
predictions_homeless_corrected = model_homeless_corrected.predict(X_homeless_corrected)

# Equation of the line for the corrected data
slope_homeless_corrected = model_homeless_corrected.coef_[0]
intercept_homeless_corrected = model_homeless_corrected.intercept_
equation_homeless_corrected = f"Homeless Total = {slope_homeless_corrected:.2f} * Year + {intercept_homeless_corrected:.2f}"

# R^2 value for the corrected data
r_squared_homeless_corrected = model_homeless_corrected.score(X_homeless_corrected, y_homeless_corrected)

# Calculate residuals for the corrected data
residuals_homeless_corrected = y_homeless_corrected - predictions_homeless_corrected

# Plotting the regression and residuals for the corrected data
plt.figure(figsize=(14, 6))

# Plot 1: Linear Regression
plt.subplot(1, 2, 1)
plt.scatter(X_homeless_corrected, y_homeless_corrected, color='blue', label='Original Data')
plt.plot(X_homeless_corrected, predictions_homeless_corrected, color='red', label='Regression Line')
plt.title('Linear Regression on Homeless Total (Corrected)')
plt.xlabel('Year')
plt.ylabel('Homeless Total')
plt.legend()
```

```python
# Plot 2: Residuals
plt.subplot(1, 2, 2)
plt.scatter(X_homeless_corrected, residuals_homeless_corrected, color='orange')
plt.axhline(y=0, color='gray', linestyle='--')
plt.title('Residuals of Linear Regression (Corrected)')
plt.xlabel('Year')
plt.ylabel('Residuals')

# Show plots
plt.tight_layout()
plt.show()

# Output the equation, R^2, and residuals for the corrected data
print("Equation of the line:", equation_homeless_corrected)
print("R^2 value:", r_squared_homeless_corrected)
print("Residuals:", residuals_homeless_corrected.tolist())
```

Seattle's Multiple Linear Regressions of features against time against the linear regression of people in poverty against time

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Data setup
years_data = np.arange(2010, 2022)
salary_data = [58566, 60183, 60624, 60856, 62196, 63945, 65489, 67490, 70152, 72478, 78776, 81245]
percentage_data = [8.8, 8.2, 7.1, 4.9, 4.4, 4, 3.9, 3.3, 3.4, 2.7, 2.2, 4.5]
population_data = [595240, 603174, 612916, 624681, 637850, 653017, 668849, 688245, 708825, 724305, 741251, 726054]
poverty_data = [451143, 459688, 414876, 432752, 450800, 439965, 428870, 417375, 398722, 374660, 350166, 346100]

# Adjusting years for the model
years = years_data - 2010

# Linear Regression for each variable
def linear_regression(x, y):
    model = LinearRegression()
    model.fit(x.reshape(-1, 1), y)
    predictions = model.predict(x.reshape(-1, 1))
    r2 = model.score(x.reshape(-1, 1), y)
    return predictions, model.coef_[0], model.intercept_, r2

# Predictions and model info
salary_predictions, salary_slope, salary_intercept, salary_r2 = linear_regression(years, salary_data)
percentage_predictions, percentage_slope, percentage_intercept, percentage_r2 = linear_regression(years, percentage_data)
population_predictions, population_slope, population_intercept, population_r2 = linear_regression(years, population_data)
poverty_predictions, poverty_slope, poverty_intercept, poverty_r2 = linear_regression(years, poverty_data)

# Plotting
plt.figure(figsize=(14, 8))

# Salary
plt.plot(years_data, salary_predictions, label=f'Salary: y={salary_slope:.2f}x+{salary_intercept:.2f} (R^2={salary_r2:.2f})', color='blue')

# Percentage
plt.plot(years_data, percentage_predictions, label=f'Percentage: y={percentage_slope:.2f}x+{percentage_intercept:.2f} (R^2={percentage_r2:.2f})', color='red')

# Population
plt.plot(years_data, population_predictions, label=f'Population: y={population_slope:.2f}x+{population_intercept:.2f} (R^2={population_r2:.2f})', color='green')

# Poverty
plt.plot(years_data, poverty_predictions, label=f'Poverty: y={poverty_slope:.2f}x+{poverty_intercept:.2f} (R^2={poverty_r2:.2f})', color='orange')

plt.xlabel('Year')
plt.ylabel('Values')
plt.title('Linear Regression Comparisons')
plt.legend()
plt.grid(True)
plt.show()
```